

Functional requirements:

Users can submit a long URL and receive a unique shortened URL.

Users can access the original URL by visiting the shortened URL.

The system should redirect users to the original URL with minimal delay.

Each generated short URL should be unique.

Non functional requirements:

The system should provide high availability (e.g., 99.99% uptime).

The system should support high scalability to handle millions of users and billions of URLs.

URL redirection should have low latency (preferably under 100 ms).

URL creation should be fast and efficient.

The system should ensure high reliability, so every short URL always redirects to the correct destination.

The system should provide data durability, ensuring URL mappings are not lost during failures.

Estimation:

Assumption that we are providing and how many users we have:

System Estimates (20M DAU)

- **Daily Writes:** 2 Million URLs/day
- **Daily Reads:** 500 Million redirects/day
- **Read / Write QPS:** ~5,800 Read QPS | ~23 Write QPS
- **Short Key Format:** 7 characters (Base62)
- **Storage (5 Years):** ~1.8 TB
- **Cache Memory (RAM):** ~50 GB

Requirements according to the estimation:

Core Requirements (20M DAU)

- **Load Balancer:** 1 Layer-7 LB (TLS termination)
- **App Servers:** 4–6 Instances (4 vCPU, 8 GB RAM each)
- **Cache:** 50 GB Redis Cluster (3 × 32 GB nodes)
- **Database:** 1 Primary DB + 2 Read Replicas (2 TB NVMe SSD)
- **Network Bandwidth:** ~50 Mbps Peak

